The logarithmic norm

$$u'(t) = Au(t) + g(t) \qquad A \in \mathbb{C}^{m\times m}, \quad u: \mathbb{R} \to \mathbb{C}^m$$

Let's derive an upper bound for $\frac{d}{dt}|u(t)|$

(note: here $|\cdot| = \|\cdot\|$ for vector norms)

$$\frac{d}{dt}|u| = \lim_{h\to 0^+} \frac{|u(t+h)| - |u(t)|}{h}$$

$$= \lim_{h\to 0^+} \frac{|u(t) + hu'(t) + \mathcal{O}(h^2)| - |u(t)|}{h}$$

$$= \lim_{h\to 0^+} \frac{|u(t) + hAu(t) + hg(t)| - |u(t)|}{h}$$

Submultiplicativity: $\|Ax\| \le \|A\| \cdot |x|$

$$\le \lim_{h\to 0^+} \frac{|u(t) + hAu(t)| - |u(t)|}{h} + |g(t)|$$

$$\le \lim_{h\to 0^+} \frac{\|I + hA\| - 1}{h} |u(t)| + |g(t)| = \mu[A]|u(t)| + |g(t)|$$

Define: $\mu[A] = \lim_{h\to 0^+} \frac{\|I + hA\| - 1}{h}$

If $g=0$: $|u(t)| \le e^{t\mu[A]}|u(0)|$. But the exact soln. is $u(t) = e^{tA}u(0)$

So $\|e^{tA}\| \le e^{t\mu[A]}$ (for $t \ge 0$) ← most important property of log. norm

and $\mu[A]$ is the smallest value such that this bound holds.

Induced matrix norm: given a vector norm $|\cdot|$, we define

$$\|A\| = \sup_{|x|=1} \frac{|Ax|}{|x|}.$$

More generally, we have (Duhamel-ish)

$$|u(t)| \leq e^{t\mu[A]}|u_0| + \int_0^t e^{(t-\tau)\mu[A]} |g(\tau)| \, d\tau$$

$$\leq e^{t\mu[A]}|u_0| + \max_{0\leq\tau\leq t}|g(\tau)| e^{t\mu[A]} \int_0^t e^{-\tau\mu[A]} d\tau$$

$$|u(t)| \leq e^{t\mu[A]}|u_0| + \max_{0\leq\tau\leq t}|g(\tau)| e^{t\mu[A]} \cdot \frac{-1}{\mu[A]}\left(e^{-t\mu[A]} - 1\right)$$

$$|u(t)| \leq e^{t\mu[A]}|u_0| + \max_{0\leq\tau\leq t}|g(\tau)| \frac{e^{t\mu[A]} - 1}{\mu[A]} \qquad (*)$$

If we apply a numerical method to

$$u'(t) = Au + g(t)$$

the global error will satisfy (as $h \to 0$)

local truncation error

$$E'(t) = AE + \tau(t)$$

Applying $(*)$ with $E(0) = 0$ we get

$$|E(T)| \leq \max_{0\leq t\leq T} |\tau(t)| \frac{e^{t\mu[A]} - 1}{\mu[A]}$$

This is much more useful than bounds based on the norm of A.

Let $A = \lambda \in \mathbb{C}$. Then $\qquad \lambda = x + iy$

$$\mu[\lambda] = \lim_{h\to 0^+} \frac{|1+h\lambda| - 1}{h} \qquad \sqrt{1+\varepsilon} = 1 + \tfrac{1}{2}\varepsilon + \mathcal{O}(\varepsilon^2)$$

$$= \lim_{h\to 0^+} \frac{\left((1+xh)^2 + (hy)^2\right)^{1/2} - 1}{h}$$

$$= \lim_{h\to 0^+} \frac{\left(1 + 2hx + \mathcal{O}(h^2)\right)^{1/2} - 1}{h}$$

$$= \lim_{h\to 0^+} \frac{1 + hx - 1}{h} = x = \mathrm{Re}(\lambda)$$

Recall: $u'(t) = \lambda u \Rightarrow |u|$ is bounded iff $\mathrm{Re}(\lambda) \le 0$

$u'(t) = Au \Rightarrow |u|$ is bounded (for general initial data) iff $\mu[A] \le 0$.

If $\mu[A] \le 0$ we say A is dissipative w.r.t. $|\cdot|$.

We can show:

$$\mu_2[A] = \max\left\{\lambda : \lambda \in \sigma\left(\frac{A+A^*}{2}\right)\right\}$$

(for Hermitian A, $\mu_2[A] = \alpha(A) = \max(\mathrm{Re}(\lambda))$, the spectral abscissa)

$$\mu_1[A] = \max_j \left(\mathrm{Re}(a_{jj}) + \sum_{i\ne j} |a_{ij}|\right)$$

$$\mu_\infty[A] = \max_i \left(\mathrm{Re}(a_{ii}) + \sum_{j\ne i} |a_{ij}|\right)$$

Example: Heat equation w/centered differences

$$u_t = u_{xx} \qquad u(0,t) = u(1,t) = 0$$
$$u(x,0) = u_0(x)$$

$u_i \approx u(x_i) = u(i\Delta x)$

$$U_i'(t) = \frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta x)^2} \Rightarrow U'(t) = AU$$

$$A = \frac{1}{(\Delta x)^2} \begin{bmatrix} -2 & 1 & & & \\ 1 & \ddots & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & -2 \end{bmatrix}$$

A is $m \times m$

$\Delta x = \frac{1}{m+1}$

We ask:

- Is the energy $\|U\|_2$ non-increasing?
- Does it satisfy the maximum principle? ($\|U\|_\infty$ non-increasing)
- Does it preserve positivity? i.e. $U(0) \geq 0 \Rightarrow U(t) \geq 0$?

Energy: is $\mu_2[A] \leq 0$?

A is Hermitian, so $\mu_2[A] = \max \lambda$.

Eigenvalues: $\lambda_j = \frac{2}{(\Delta x)^2}(\cos(j\pi\Delta x) - 1) \qquad j = 1, \dots, m$

Recall: $\cos x = 1 - \frac{1}{2}x^2 + \mathcal{O}(x^4)$

So $\lambda_j = -j^2\pi^2 + \mathcal{O}(\Delta x^2)$ max occurs for $j=1$: $\lambda_1 \approx -\pi^2$

So $\mu_2[A] = -\pi^2 + \mathcal{O}(\Delta x^4)$

and

$$\|u(t)\|_2 \leq \|e^{tA}\|_2 \|u(0)\|_2 \leq e^{-\pi^2 t} \|u(0)\|$$

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}$$

$$u'(t) = Au(t)$$
$$u_1(t) = e^{2t} u_1(0)$$
$$u_2(t) = e^{-5t} u_2(0)$$

$$\mu_2[A] = 2$$

What if we take $\lim\limits_{h \to 0} \dfrac{\|I + hA\| - 1}{h} = -\mu[-A] = -5$